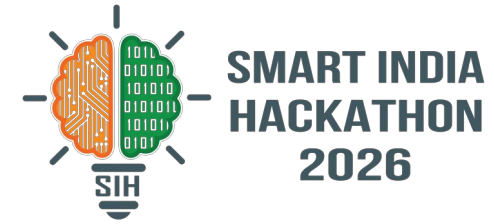
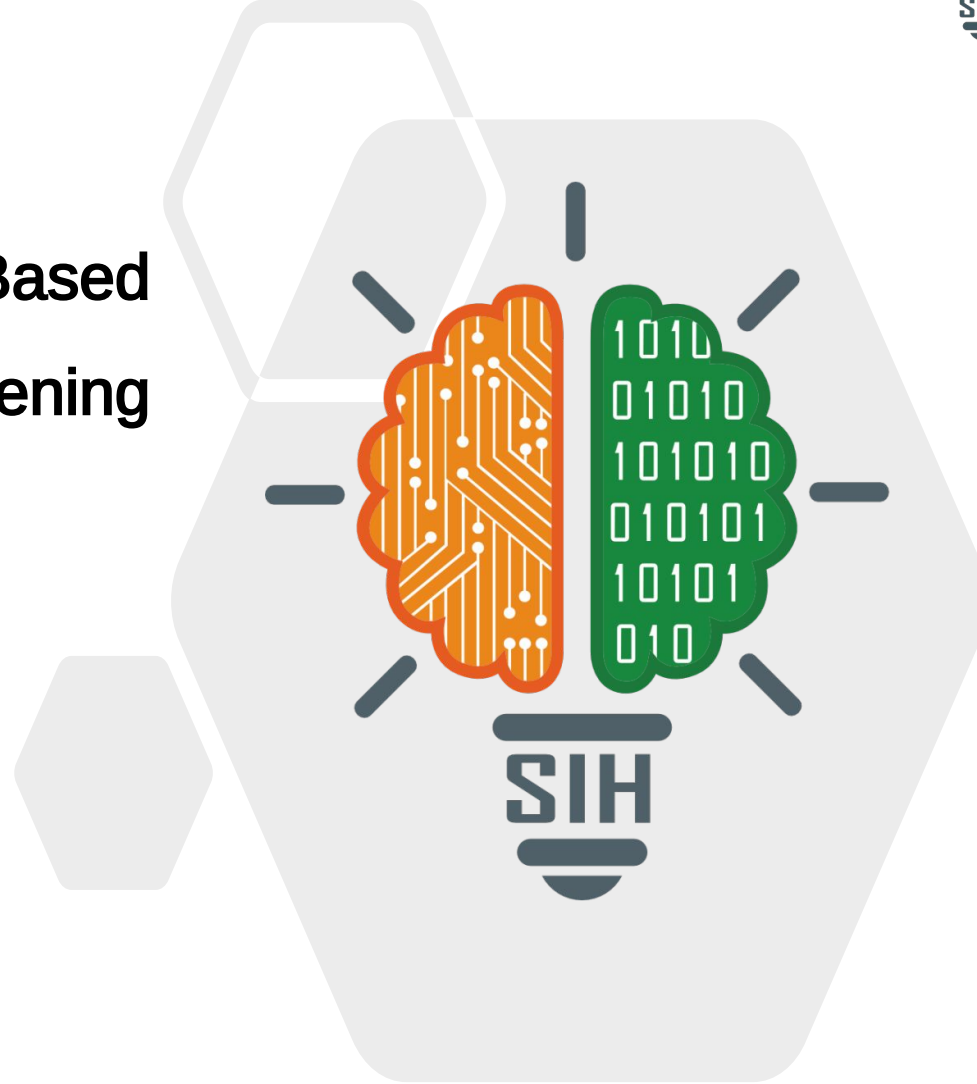


SMART INDIA HACKATHON 2026



- Problem Statement ID – SIH26188
- Problem Statement Title- AI- Based Fake Identity and Document Screening System
- Theme- Miscellaneous
- PS Category- Software
- Team ID-SIH262
- Team Name- INNOVEXA



❖ Proposed Solution

PehchaanScan — Instant Document Screening for Border Security

REAL-WORLD ISSUE

Border posts check thousands of passports, visas & IDs daily - by eye. Slow queues. Clever fakes slip through: swapped photos, edited dates, forged stamps, invented numbers.

WHY IT MATTERS

One missed fake = one breach.
Verification: minutes per person.

OUR SOLUTION

Photograph any document -> machine second-opinion in 5 seconds: read, validate, tamper-scan, face-match, risk score. Officer decides.

4-MODULE SCREENING ENGINE

1. OCR READ

Extract name, number, DOB, expiry

2. VALIDATE

Format + expiry + Aadhaar checksum

3. TAMPER SCAN

ELA heatmap on photo + metadata

4. FACE MATCH

Document photo vs live selfie

RISK SCORE

GREEN - CLEAR

AMBER - CHECK

RED - DETAIN

THREAT -> ANSWER

Invented ID numbers

Checksum: 100% catch

Photo replacement

ELA scan on photo

Impersonation

Live face match

Edited text / dates

Format & logic checks

Long queues

5-second screening

WHAT MAKES IT UNIQUE

MATH, NOT GUESSWORK

Verhoeff + MRZ check digits:
fake numbers fail the math

FORENSICS THAT COUNT

ELA aimed at the photo region
- the #1 forgery spot

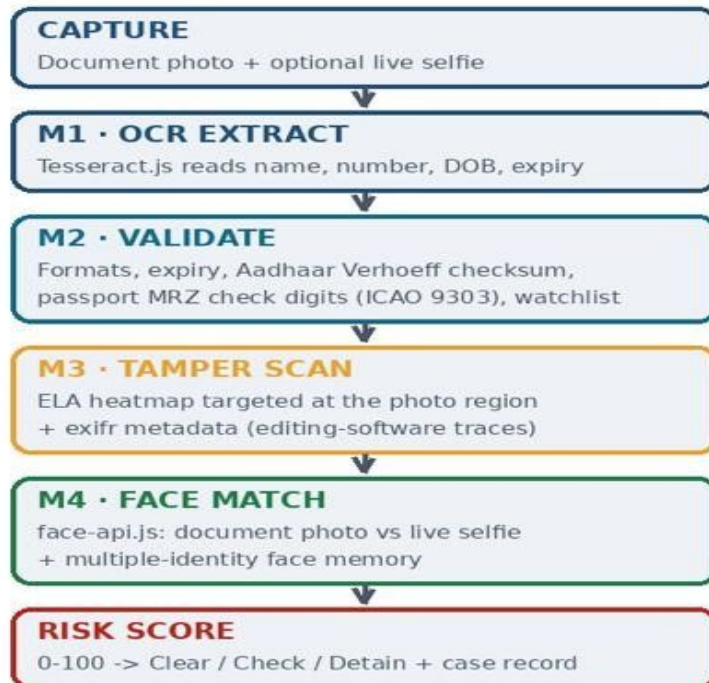
PRIVACY BY DESIGN

100% on-device - documents
never leave the phone

ASSISTS, NOT REPLACES

Machine flags, officer decides
- case ID on every scan

Technical Approach — all four modules working, fully on-device



TECHNOLOGIES USED



Hardware needed: just a smartphone camera.

Feasibility & Viability — execution plan, honest risks, field survival

CAN WE BUILD IT?

Standards are public

Aadhaar checksum & passport MRZ are published specs - validation is exact, not an AI guess anyone can dispute

Already built

Working prototype covers all four modules + watchlist and multi-identity detection

Develops without real IDs

Team tests on specimen documents - no privacy clearance needed to build or to demo on stage

Zero running cost

No cloud, no API bills - identical cost at 10 scans or 10 lakh scans

RISKS -> MITIGATIONS

Blur & bad light at busy posts

Quality gate rejects unusable photos instantly with a retake prompt - under 3 seconds lost

Regional-language documents

v1 reads the English-side fields present on all major IDs; language packs are a config add, not a rebuild

Tamper signals can false-alarm

Shown as 'flag for review', never auto-reject; officer override is one tap - and every override logged

Officer adoption

One screen, zero typing - training takes five minutes

WILL IT SURVIVE THE FIELD?

Deployable today

No hardware purchase, no server procurement - a web link opened on phones officers already carry

Fits the SOP

Assists, never replaces: final call stays human, audit trail keeps accountability where it belongs

Grows sideways

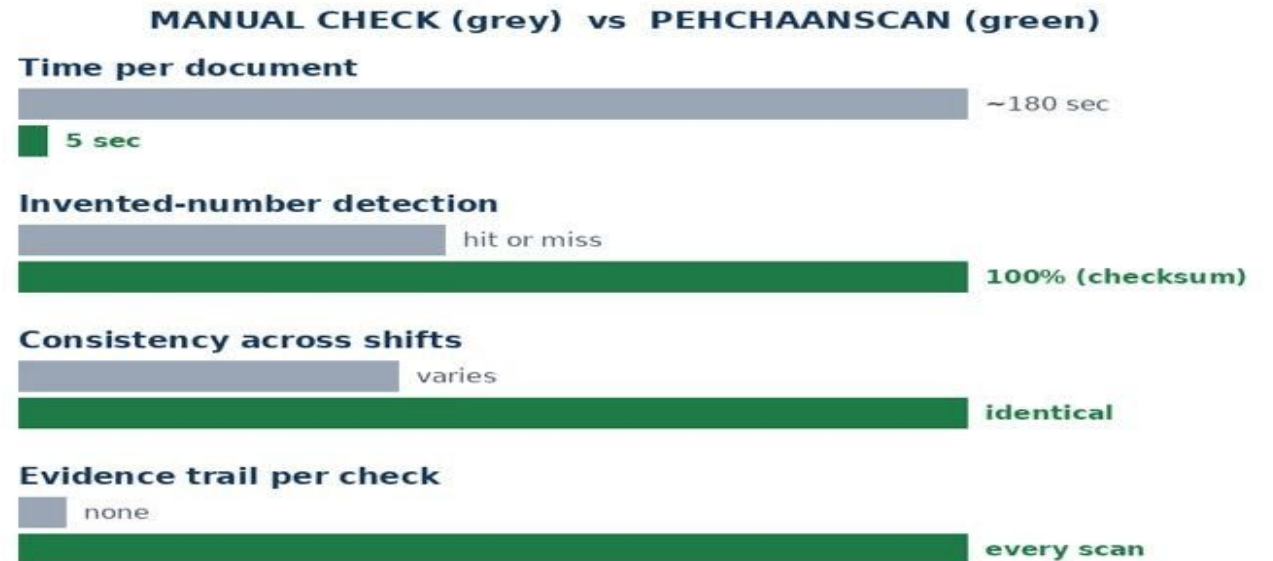
Same engine, new checklist: bank KYC, SIM issue, exam halls, hotel check-in

Grows upward

Security-grade AI models & UIDAI QR verification plug in; rule updates ship as config when standards change

Impact & Benefits — from gut feeling to machine-checked, in seconds

- 1 **OFFICERS**
Minutes -> seconds per document;
uniform standard at every post
- 2 **SECURITY**
Invented numbers caught by math;
photo-swap & edit flags every scan
- 3 **TRAVELLERS**
Shorter queues - honest passengers
move faster through checkpoints
- 4 **INVESTIGATIONS**
Every scan logged: digital trail
feeds intelligence & case files
- 5 **PRIVACY**
Zero uploads - documents processed
and forgotten on the device



SAME ENGINE, MANY DOORS

BANK KYC

SIM ISSUANCE

HOTEL CHECK-IN

EXAM-HALL ID

GOVT OFFICE ENTRY

One configurable screening engine wherever identity meets a queue.



	UIDAI (Aadhaar number structure & checksum) https://uidai.gov.in
	ICAO Doc 9303 (passport MRZ & check-digit standard) https://www.icao.int
	Sashastra Seema Bal / MHA (problem owner) https://ssb.gov.in
	Tesseract.js (OCR) https://github.com/naptha/tesseract.js
	face-api.js (face matching) https://github.com/justadudewhohacks/face-api.js
	exifr (metadata analysis) https://github.com/MikeKovarik/exifr
	Error Level Analysis reference https://fotoforensics.com

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- Tesseract.js (OCR) — <https://github.com/naptha/tesseract.js>
- face-api.js (face matching) — <https://github.com/justadudewhohacks/face-api.js>
- exifr (metadata analysis) — <https://github.com/MikeKovarik/exifr>
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